MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155 MM, XM712, THERMAL BATTERY FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for manufacture and acceptance of the XM712 Guided Projectile thermal battery, critical item.

2. APPLICABLE DOCUMENTS

2.1 Government documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	-	Box, Wood, Cleated, Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-850	-	Cushioning Material, Polystyrene, Expanded, Resilient (for Packaging Applications)
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular, Polyethylene Foam, Flexible
PPP-T-76	-	Tape, Packaging, Paper (for Carton Sealing)

Military

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaged Use and Static Dehumidification
MIL-P-17667	-	Paper, Wrapping, Chemically Neutral (Noncorrosive)

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-810 - Environmental Test Methods

DRAWINGS

Army

9301200 - Battery, Thermal, Guided Projectile XM712

OTHER PUBLICATIONS

Regulations

AR 700-15 - Packaging of Material

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- The thermal battery for the XM712 Cannon Launched Guided Projectile, is a primary component of the control section and is mounted on the control actuator assembly. The battery provides power output of +11, +30 and -30 volts direct current (Vdc). The +11 Vdc section is activated by an inertial starter mechanism and the ± 30 Vdc sections are activated by an electrical initiator. The electrical interface is made through terminals located at each end of the battery.

3.2 CHARACTERISTICS

3.2.1 Performance.- The thermal battery shall comply with the performance requirements of this specification under the operational environment specified herein after being subjected to the non-operational and cannon launch environments specified herein.

3.2.1.1 Electrical requirements.- The thermal battery shall provide output power at +11, +30, and -30 Vdc when loaded under maximum or minimum loads and operated under the conditions as specified herein with the +11 volt and the ± 30 volt common terminals electrically isolated from the battery case.

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3.2.1.1.1 +11 Volt source.- The +11 volt source output voltage and the applied load profiles shall comply with the requirements of Figures 1 and 2. The +11 volt output voltage shall have a minimum lifetime of 100 seconds.

3.2.1.1.1.1 +11 Volt squib load.- There shall be four (4) squib loads applied to the +11 volt source as specified in Table I.

3.2.1.1.2 +30 Volt source.- The +30 volt source output voltage and the applied load profile shall comply with the requirements of Figures 3 and 4. The +30 volt source shall have a minimum lifetime of seventy (70) seconds.

3.2.1.1.2.1 +30 Volt source squib loads.- There shall be five 4.75 ± 0.68 ohm squib loads applied to the +30 volt source. The squib loads shall occur at 1, 2, 3, 4, and 5 seconds after initiation and shall have a duration of 20 ± 1 milliseconds. The +30 volt source output voltage shall be +30.0 volts minimum during applied squib loads and shall recover to within +0,-2 volts, of the same voltage level as that prior to the application of each of the respective squib loads, in a maximum of 75 milliseconds.

3.2.1.1.3 -30 Volt source.- The -30 volt source output voltage and the applied load profiles shall comply with the requirements of Figures 5 and 6. The -30 volt output voltage shall have a minimum lifetime of seventy (70) seconds.

3.2.1.1.4 Voltage variations (noise).- No voltage output source shall exhibit voltage variations, other than those due to load changes and source impedance changes during activation, except to the extent specified herein. The amplitude of any voltage variation shall not exceed three (3) volts peak to peak for the first 1.5 seconds after the section has attained minimum specified voltage. From the first 1.5 seconds to the end of specified life, voltage variations occurring more than once per second shall not exceed 0.35 volts peak to peak.

3.2.1.2 Activation requirements.-

3.2.1.2.1 +11 Volt source activation.- The +11 volt source is activated by an inertial starter which shall comply with the following requirements.

3.2.1.2.1.1 No-Fire stimulus.- The inertial starter no-fire stimulus range shall be 0 to 490 g acceleration for a time period of 4 milliseconds (1/2 sine pulse).

3.2.1.2.1.2 All-Fire stimulus.- The inertial starter minimum all-fire stimulus range shall be 1000 ± 100g acceleration for a time period of 4 ± 1 milliseconds, (1/2 sine pulse).

3.2.1.2.2 ± 30 Volt source activation.- The ± 30 volt source is activated by an electrical match initiator which shall comply with the following requirements.

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3.2.1.2.2.1 Bridge resistance.- The electrical initiator shall have a bridge resistance of 0.25 ± 0.15 ohms at 50 milliamperes maximum, prior to activation. After activation and during specified life, the bridge circuit internal to the battery shall have a maximum leakage current of 100 milliamperes to all output terminals and case of the ± 30 volt section of the battery.

3.2.1.2.2.2 No-fire current.- The electrical initiator no-fire current range shall be from zero (0) up to and including one (1) ampere dc for a maximum of 15 seconds at 69°F (21°C).

3.2.1.2.2.3 All-fire current.- The electrical initiator all-fire current range shall be ≥ 4.5 amperes dc for a minimum of 20 milliseconds.

3.2.1.3 Activation time requirements.-

3.2.1.3.1 +11 Volt source activation time.- The +11 volt source output voltage activation time shall fall within the voltage versus time boundaries specified in Figure 7.

3.2.1.3.2 +30 Volt source activation time.- The +30 volt source output voltage activation time shall fall within the voltage versus time boundaries specified in Figure 8.

3.2.1.3.3 -30 Volt source activation time.- The -30 volt source output voltage activation time shall fall within the voltage versus time boundaries specified in Figure 9.

3.2.1.4 Terminal strength.- Each battery terminal shall withstand a pull test without rupture or damage under an applied force of four pounds maximum.

3.2.1.5 Resistance to soldering heat.- No damage or degradation of performance shall occur after a 5 second solder application at +600°Fahrenheit (F) (315.5° (C)) to any terminal at a point ≥ 0.12 inch from the case.

3.2.1.6 Insulation resistance.- The insulation resistance between the case and any of the power output terminals and between the case and the shorted electrical initiator terminals shall be 5 megohms minimum when a minimum of 500 Vdc is applied.

3.2.1.7 Dielectric resistance.- The dielectric resistance between any terminal and any other terminal including the shorted electrical initiator terminals shall be 5 megohms minimum when a minimum of 500 Vdc is applied.

3.2.1.8 Case leakage.- The thermal battery shall exhibit no evidence of leakage from either section before and during the activated life of the battery.

3.2.1.9 Polarity.- The battery polarity shall comply with the requirements of 3.2.1.1.

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3.2.2 Environmental conditions.- The thermal battery shall meet the performance requirements of 3.2.1 during and/or after exposure to the following environmental conditions, as specified.

3.2.2.1 High temperature.- The thermal battery shall meet the requirements of 3.2.1 following exposure to ambient temperatures of 160° Fahrenheit (F) (71°Celcius (C)) and during ambient temperatures of 145°F (62.8°C).

3.2.2.2 Low temperature.- The thermal battery shall meet the requirements of 3.2.1 following exposure to ambient temperatures of -50°F (-45.6°C) and during ambient temperatures of -25°F (-31.7°C).

3.2.2.3 Temperature shock.- The thermal battery shall meet the requirements of 3.2.1 following exposure to temperature shock in either direction between -50°F (-45.6°C) and 160°F (71°C).

3.2.2.4 Launch shock.- The thermal battery shall meet the requirements of 3.2.1 following exposure to a forward longitudinal acceleration of 9000 ± 300 g, 1/2 sine pulse, 10 millisecond minimum duration.

3.2.2.5 Overpressure.- The thermal battery shall meet the requirements of 3.2.1 following exposure to a peak pressure of 600 psi for a minimum of 6.0 milliseconds (ms).

3.2.2.6 Transportation vibration.- The thermal battery shall meet the requirements of 3.2.1 following exposure to sinusoidally cycled vibrations ranging from 2.5 g's double amplitude at 5 Hz to 92 Hz, 0.0058 in. double amplitude at 93 Hz to 110 Hz and 3.5 g's double amplitude at 111 to 200 Hz including sinusoidal dwell vibrations of ± 20 g's and ± 35 g's at 110 Hz.

3.2.2.7 Fragility shock.- The thermal battery shall withstand exposure to half sine wave mechanical shock pulses having nominal durations of 10 ms with maximum peak accelerations of 200 g's.

3.3 Design and Construction.-

3.3.1 Production drawings.- The thermal battery shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on drawing 9301200.

3.3.2 Workmanship.- All parts shall be fabricated and finished in a throughly workmanlike manner. Except as otherwise specified, the parts shall be free of cracks, scratches, burrs, dents, gouges, surface defects or any surface discontinuities. The items shall be free from dirt, chips, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

3.4 First article inspection.- This specification makes provisions for first article inspection. First article samples shall meet all the requirements of this specification. Requirements for submission of the first article samples by the contractor shall be as specified in the contract (See 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or procurement document, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as specified herein, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The following types of inspections shall be conducted on this item:

(a) First article inspection (4.3).

(b) Quality conformance inspection (4.4).

4.3 First article inspection.-

4.3.1 Submission.- The contractor, when required by contract, shall submit a first article sample of 12 thermal batteries for examination and test by procuring activity in accordance with Table II. The first article samples shall have been inspected for the characteristics listed in Table III. Environmental tests shall be conducted in the sequence indicated. The operating performance of each thermal battery shall be as indicated by each test method. The testing shall be considered destructive. All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and material shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the procuring activity. A first article sample, or portion thereof, as directed by the procuring activity, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification, such as to significantly affect product uniformity, as determined by the procuring activity.

4.3.2 Inspection to be performed.- First article test specimens may be subjected by the procuring activity to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings.

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4.3.3 Rejection.- If any test specimen fails to comply with any of the applicable requirements, the first article samples shall be rejected. The procuring activity reserves the right to terminate its inspection upon the failure of any test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.-

4.4.1 Inspection lot.- Unless otherwise specified herein, or by the procuring activity, inspection lot size, lot formation, and presentation of lots, shall be in accordance with the inspection and sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be no greater than one month's production. Each lot of thermal batteries shall contain inertial starters and initiators from not more than two (2) lot numbers of the same configuration from one manufacturer. All other parts shall be of one configuration.

4.4.2 Inspections.-

4.4.2.1 Sampling inspection.- Each lot thermal batteries shall consist of hardware which has been inspected per Table III. Unless otherwise specified, sampling shall be to level II of MIL-STD-105. Lots failing to meet the requirements of Table III shall be dispositioned in accordance with the applicable requirements of MIL-STD-105.

4.4.2.2 Lot acceptance testing.- Each lot of thermal batteries shall be inspected in accordance with Table IV. Sample size shall be in accordance with MIL-STD-105, level S-3, AQL 1.0%. Lots failing to meet these requirements shall be rejected.

4.5 Test methods.-

4.5.1 Test conditions.-

4.5.1.1 Environmental conditions.- Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

(a)	Temperature	72° ± 18°F
(b)	Relative Humidity	90 percent maximum
(c)	Barometric Pressure	28 to 32 inches of mercury

4.5.1.2 Test tolerances.- All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

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4.5.1.3 Inspection equipment.- The inspection equipments required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the inspection tables included herein. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (see 6.4).

4.5.2 Performance tests.-

4.5.2.1 Terminal strength.- Test terminal strength as specified in MIL-STD-202, Method 211, Test Condition A, using a four pound force, maximum. After test, visually inspect terminals for rupture or damage and for compliance with the requirements of 3.2.1.4.

4.5.2.2 Resistance to soldering heat.- Adjust soldering iron temperature to 600°F-615°F (316°C-324°C) and wet with Sn 60 solder. Apply solder iron tip to terminal no less than 0.12 inch from battery for 5 seconds minimum. After test, visually inspect terminals for damage and compliance to requirements of 3.2.1.5.

4.5.2.3 Case leakage.- The battery shall be subjected to a minimum pressure of 80 psig nitrogen for a minimum of 30 minutes. Within three minutes of removal from the pressure chamber, the battery shall be immersed in a hot water bath at 160°F ± 10°F (71°C ± 5.6°C) for a minimum of 5 minutes. The battery shall not emanate bubbles or show evidence of leakage and shall comply with the requirements of 3.2.1.8.

4.5.2.4 Insulation resistance.- Using an appropriate megohmmeter, measure insulation resistance at 500 Vdc between case and each power output terminal and between case and shorted electrical initiator terminals. The resistance shall meet the requirements of 3.2.1.6.

4.5.2.5 Dielectric resistance.- Using an appropriate megohmmeter, measure dielectric resistance at 500 Vdc between each terminal and each other terminal including the shorted electrical initiator terminals. The resistance shall meet the requirements of 3.2.1.7.

4.5.2.6 ± 30 Volt source bridge resistance.- Measure the resistance across the two match "M" terminals using an ohmmeter at a maximum current of 10 milliamperes. The match resistance shall meet the requirements of 3.2.1.2.2.1.

4.5.2.7 ± 30 Volt source match no-fire.- Apply a direct current of 1.0 ampere, maximum, across the match "M" terminals through the electrical initiator for 15 seconds, maximum, using a suitable power source. The battery shall meet the requirements of 3.2.1.2.2.2.

4.5.2.8 +11 Volt source no-fire.- The battery shall be mounted to an appropriate shock machine and the no-fire stimulus of 3.2.1.2.1.1 applied to the battery. The +11 volt section of the battery shall not be activated. If during performance of the no-fire test either the 490 g level or the 4 millisecond duration are exceeded, the test item shall be considered destructed and unacceptable for any further testing.

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4.5.2.9 Functional tests.-

4.5.2.9.1 Activation test.- The battery shall be mounted to an appropriate shock machine and the 11 volt section activated by applying the all-fire stimulus of 3.2.1.2.1.2. The ± 30 volt section of the battery shall be activated by applying the 11 volt section output across the ± 30 volt section electrical initiator at 200 milliseconds after the 11 volt section output voltage reaches 10 ± 0.25 volts for a duration of 20 ± 1 milliseconds. The 11 volt section shall meet the requirements of 3.2.1.2.1.2 and 3.2.1.3.1. The ± 30 volt section shall meet the requirements of 3.2.1.2.2.3, 3.2.1.3.2, and 3.2.1.3.3.

4.5.2.9.2 Load/life test.- The battery shall be loaded in accordance with Figures 1 through 6. Voltage, current, noise, and match leakage current shall be continuously recorded throughout the mission life of each section of the battery. One-third (1/3), or the next lowest integer, of the units to be tested shall be tested to the minimum load conditions and the remaining units shall be tested at the maximum load conditions. The shock pulse of 4.5.2.9.1 shall have been recorded and its application shall be the time zero reference for the 11 volt section. Time zero for the ± 30 volt sections shall be application of the ± 30 volt all fire squib pulse. The battery shall meet the requirements of 3.2.1.1.1, 3.2.1.1.1.1, 3.2.1.1.2, 3.2.1.1.2.1, 3.2.1.1.3, and 3.2.1.1.4.

4.5.2.10 Polarity test.- Measure battery polarity with an electrometer with a minimum input resistance of 10^{14} ohms. The polarity shall be as indicated by the terminal markings and shall meet the requirements of 3.2.1.9. If a proper indication cannot be obtained at room ambient temperature, the battery shall be conditioned at 160°F ± 10°F (71°C ± 5.6°C) for one (1) hour minimum, and re-tested.

4.5.3 Environmental tests.-

4.5.3.1 High temperature operational test.- The thermal battery shall be subjected to the high temperature tests of MIL-STD-810, Method 501.1, Procedure I, Steps 1 through 5. The operating temperature shall be 145° ± 2.5°F (62.7° ± 1.4°C). The test shall be performed in the temperature conditioning chamber or within 5 minutes after removal from temperature conditioning. During operation of the thermal battery, it shall meet the requirements of 4.5.2.9.

4.5.3.2 Low temperature operational test.- The thermal battery shall be subjected to the low temperature tests of MIL-STD-810, Method 502.1, Procedure I, Steps 1 through 5, with the exception that the low storage temperature of Step 2 shall be -50° + 2.5°F (-45.6 + 1.4°C). The operating temperature of Step 4 shall be -25° ± 2.5°F (-31°F ± 1.4°C). The test shall be performed in the temperature conditioning chamber or within 5 minutes after removal from temperature conditioning. During operation of the thermal battery it shall meet the requirements of 4.5.2.9.

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4.5.3.3 Temperature shock test.- The thermal battery shall be subjected to the temperature shock tests of MIL-STD-810, Method 503.1, Procedure I, with the exception that the low temperature of Step 2 shall be -50° + 2.5°F (-45.6° + 1.4°C). The thermal battery shall meet the requirements of 3.2.1.2.1.1, 3.2.1.2.2.1, 3.2.1.2.2.2, 3.2.1.6, and 3.2.1.7, following this exposure.

4.5.3.4 Overpressure test.- Place the thermal battery in an appropriate pressure test chamber at room ambient temperature and increase the internal pressure of the chamber to 600 psi minimum. Maintain pressure for six (6) milliseconds minimum. Reduce pressure to room ambient pressure. The battery shall meet the requirements of 4.5.3.1 or 4.5.3.2 following this exposure.

4.5.3.5 Transporation vibration test.- Affix the thermal battery to a vibration table in a manner which will simulate its normal mounting. Subject the thermal battery to the vibration tests of MIL-STD-810, Method 514.2, Procedure X using the criteria of Table 514.2-VII and Figure 514.2-7 curve AX, except that dwell shall be as follows, 20 g's double amplitude at 110 Hz for 5 minutes in each of 3 axes and 35 g's double amplitude at 110 Hz for 80 seconds in both lateral axes. Following exposure to this environment the thermal battery shall meet the requirements of 3.2.1.2.1.1, 3.2.1.2.2.1, 3.2.1.2.2.2, 3.2.1.6, and 3.2.1.7.

4.5.3.6 Fragility shock test.- With the thermal battery supported in a manner which simulates its normal installation mounting, subject the battery to the shock test of MIL-STD-810, Method 516.2, Procedure I, using the shock pulse of Figure 516.2-2 with the exception that peak acceleration "A" shall be 200g and nominal duration "D" shall be 10 ms. After exposure, the thermal battery shall be examined and show no evidence of physical damage and shall meet the performance requirements of 3.2.1.2.1.1, 3.2.1.2.2.1, 3.2.1.2.2.2, 3.2.1.6, and 3.2.1.7.

4.5.3.7 Launch shock, high g test.- The thermal battery shall be installed in an appropriate canister in a manner that will facilitate activation of the 11 V section, and the assembly shall be exposed to the launch shock of 3.2.2.4, from an 8 inch gun, M115 or equivalent (GFE), at 70°F ± 30°F (21.1°C ± 16.7°C). After exposure, the thermal battery shall be examined and shall show no evidence of physical damage. The 11 V section shall have been activated by this exposure (3.2.1.2.1.2) and the ± 30 V sections shall meet the performance requirements of 4.5.2.9 for the first 20 seconds of life, at minimum load conditions, and at room ambient temperature, following this exposure. The battery shall meet the requirements of 3.2.1.8 throughout this test.

4.5.3.8 Launch shock, low g test.- The thermal battery shall be installed in an appropriate canister in a manner that will facilitate activation of the 11 V section and the assembly exposed to a launch shock of 800 g ± 200g, 1/2 sine pulse, 10 millisecond minimum duration, from an 8 inch gun, M115 or equivalent (GFE), at 70°F ± 30°F (21.1°C ± 16.7°C). After exposure, the thermal battery shall be examined and shall show no evidence of physical

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damage. The 11 V section shall have been activated by this exposure (3.2.1.2.1.2) and the ± 30 V sections shall meet the performance requirements of 4.5.2.9 for the first 20 seconds of life, at minimum load conditions, and at room ambient temperature, following this exposure. The battery shall meet the requirements of 3.2.1.8 throughout this test.

4.5.4 Physical dimensions.- Using appropriate gages and inspection, verify the requirements of 3.3.1.

4.5.5 Workmanship.- Visually inspect welding area and battery for voids, scratches, dents, nicks, and corrosion. Verify that battery, terminals, and terminal identification comply with drawing 9301200 and that workmanship in general complies with the requirements of 3.3.2.

4.5.6 Preparation for delivery inspection.- The preservation, packaging, packing and marking of the thermal battery for shipping shall be inspected for compliance with the requirements of 5.0.

5. PREPARATION FOR DELIVERY

5.1 General.- Packaging and packing shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Preparation.-

5.1.1.1.1 Terminal protection.- All electrical battery terminals shall be coated with a protective agent.

5.1.1.1.2 Battery cleaning.- The battery shall be cleaned of all foreign matter prior to packaging as required. The cleaning solvent shall not degrade the battery finish or cause any damage to the battery.

5.1.1.1.3 ± 30 Volt shunt.- A shunt shall be placed across the two "M" terminals of the ± 30 volt source electrical initiator.

5.1.1.1.4 Terminal caps.- Terminal caps shall be placed over both ends of the battery. These caps shall be capable of preventing damage to the terminals and threads of the battery.

5.1.1.2 Packaging.- The thermal battery shall be packaged in accordance with AR700-15, Level B. Sharp edges and protrusions shall be covered with PPP-C-843 cushioning material. The assembly shall then be wrapped with Type II, MIL-P-17667, barrier material. Desiccant conforming to MIL-D-3464 shall be spaced uniformly around the battery and secured with PPP-T-76 tape. The battery shall then be heat-sealed in a Class F, MIL-B-117 bag.

5.1.1.3 Packing.- The packaged thermal battery shall be cushioned within the exterior container with polystyrene or polyethylene pad and saddles, PPP-C-850 or PPP-C-1752 respectively. The exterior container shall conform to PPP-B-601 or PPP-B-621, overseas requirements.

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5.1.2 Level B.-

5.1.2.1 Preparation.- Same as Level A.

5.1.2.2 Packaging.- The thermal battery shall be packaged in accordance with MIL-P-116, Method IA-8. Sharp edges and protrusions shall be cushioned with PPP-C-843. The assembly shall then be heat sealed within a Class F, MIL-B-117 bag.

5.1.2.3 Packing.- Packing shall be accomplished as defined in 5.1.1.3 with the exception that the exterior container shall be a domestic type.

5.1.3 Commercial.-

5.1.3.1 Preparation.- Same as Level A.

5.1.3.2 Packaging.- The thermal battery shall be packaged in a manner that will prevent contamination of the item from the supply source to the first receiving activity for immediate use.

5.1.3.3 Packing.- The thermal battery shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with uniform freight classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.2 Marking.- Unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and shall show any special marking required by contract or order.

6. NOTES

6.1 Intended use.- The thermal battery is intended for use in the XM712 Projectile.

6.2 Ordering data.- Procurement documents shall specify the following:

6.2.1 Procurement Requirements.-

(a) Title, number, and date of this specification.

(b) Quantity required and delivery schedules.

(c) Serialization requirements, if applicable.

(d) Quality Conformance Inspection, if other than specified in Section 4 of this specification.

(e) First Article Sample requirements, if other than specified in Section 4 of this specification.

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(f) Packaging requirements, if other than specified in Section 5 of this specification.

(g) Certificates of conformance for each lot or shipment of product.

(h) Requirements for lot control.

6.2.2 Contract data requirements.-

(a) Contract data requirements for inspection equipment designs (conforming to data Item description DI-E-1118 or DI-R-1714 with appropriate addendum to each).

(b) Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions.-

6.3.1 Contractor.- Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned, Contractor Operated (GOCO) plants.

6.3.2 Technical Data Package (TDP).- A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists of specifications, standards, models, performance requirements, quality assurance provisions, and packaging data.

6.3.3 Inspection.- The contractor will not assume, nor does the Government guarantee that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerance and fit within the limits of the specification and drawings that best suit his process needs and still satisfy the testing requirements.

6.4 Inspection equipment designs.- Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Tables II, III, and IV, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless

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otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE Designs - SIE designs may consist of any of the following:

(a) Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

(b) A source control drawing or a specification control drawing as defined in MIL-STD-100.

(c) An envelope drawing, as defined in MIL-STD-100,which establishes the criteria which a detailed design shall meet. When the envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor designs.- Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

6.4.3 Submission of designs for approval.- Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distributions specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. The specified segment of Picatinny Arsenal to which the contractor designs shall be sent will be specified in the item detail specification.

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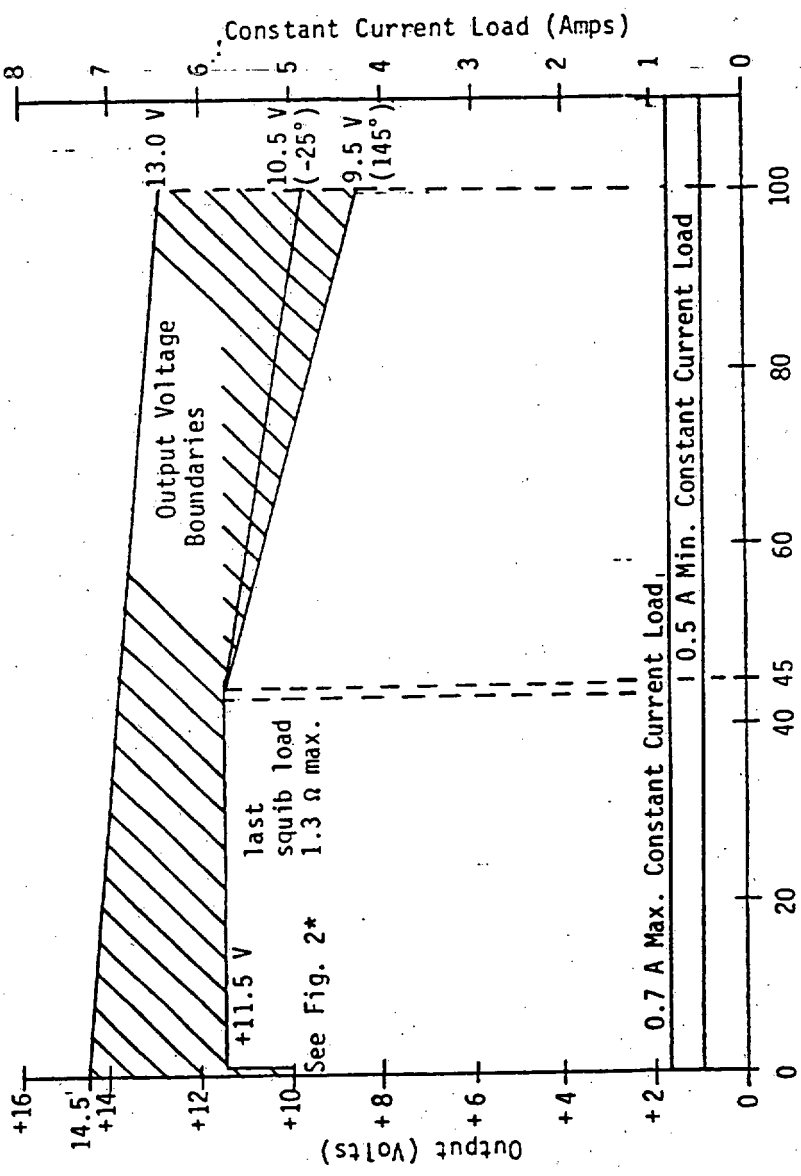


Figure 1. +11 Volt Power Source Voltage and Load Profiles (0 to 100 Seconds)

*Refer to paragraph 3.2.1.1.1.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

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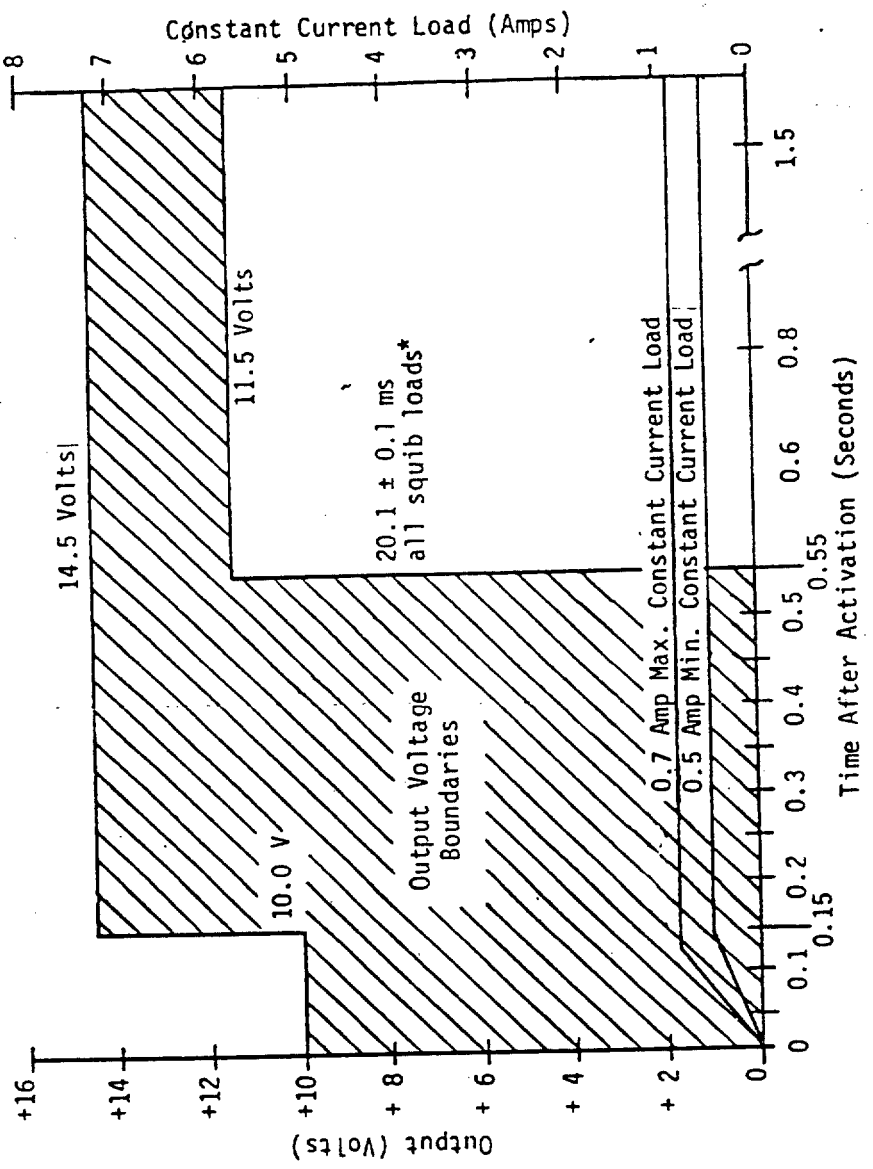


Figure 2. +11 V Power Source Voltage and Load Profiles (0 to 1.5 seconds)

*Refer to paragraph 3.2.1.1.1.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

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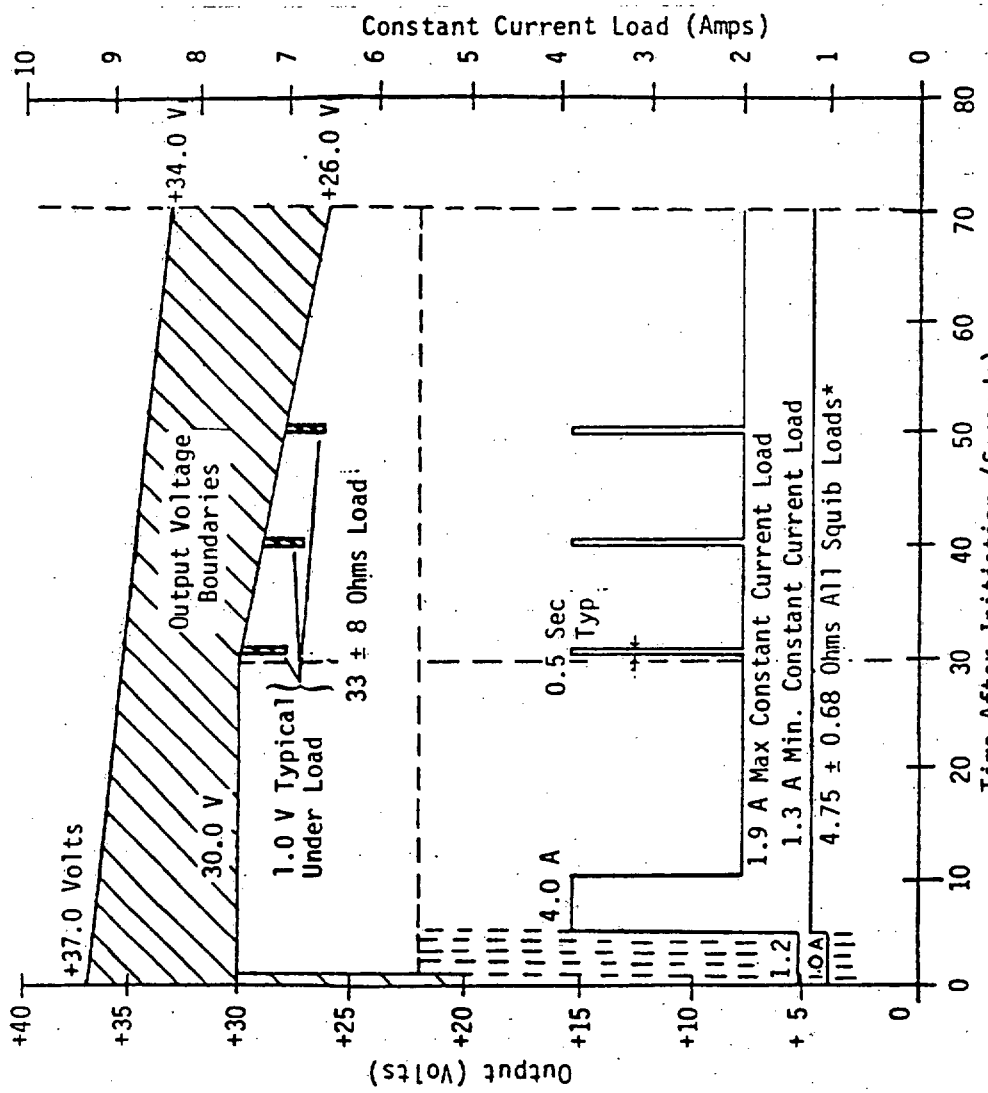


Figure 3. +30 Volt Power Source Voltage and Load Profiles (0 to 70 Seconds)

*Refer to paragraph 3.2.1.1.2.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

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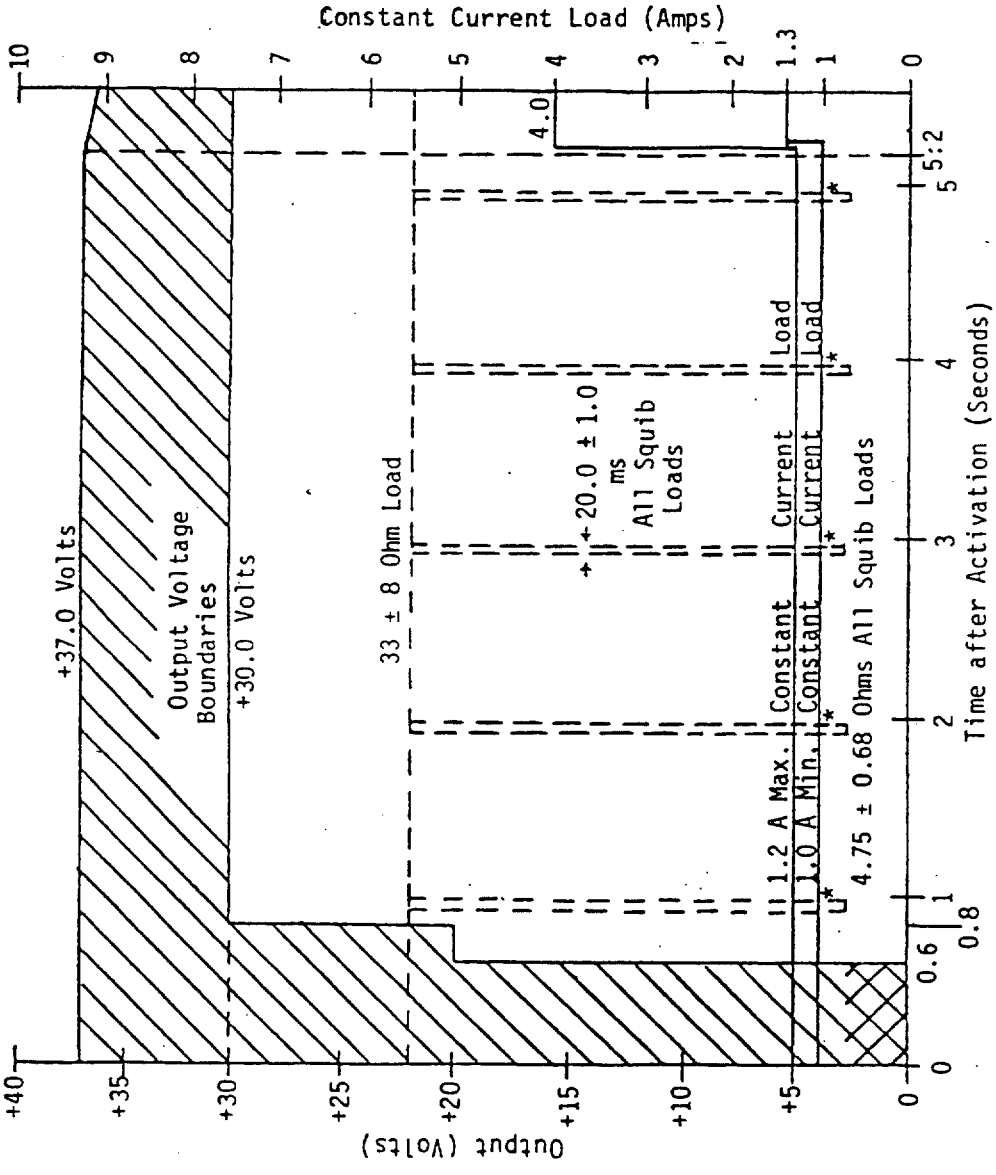


Figure 4. +30 Volt Power Source Voltage and Load Profiles (0 to 5.5 Seconds)
(Typical Squib Load Pattern Shown)

*Refer to paragraph 3.2.1.1.2.1 for times at which squib loads can occur, minimum voltage permissable during squib loads and recovery time.

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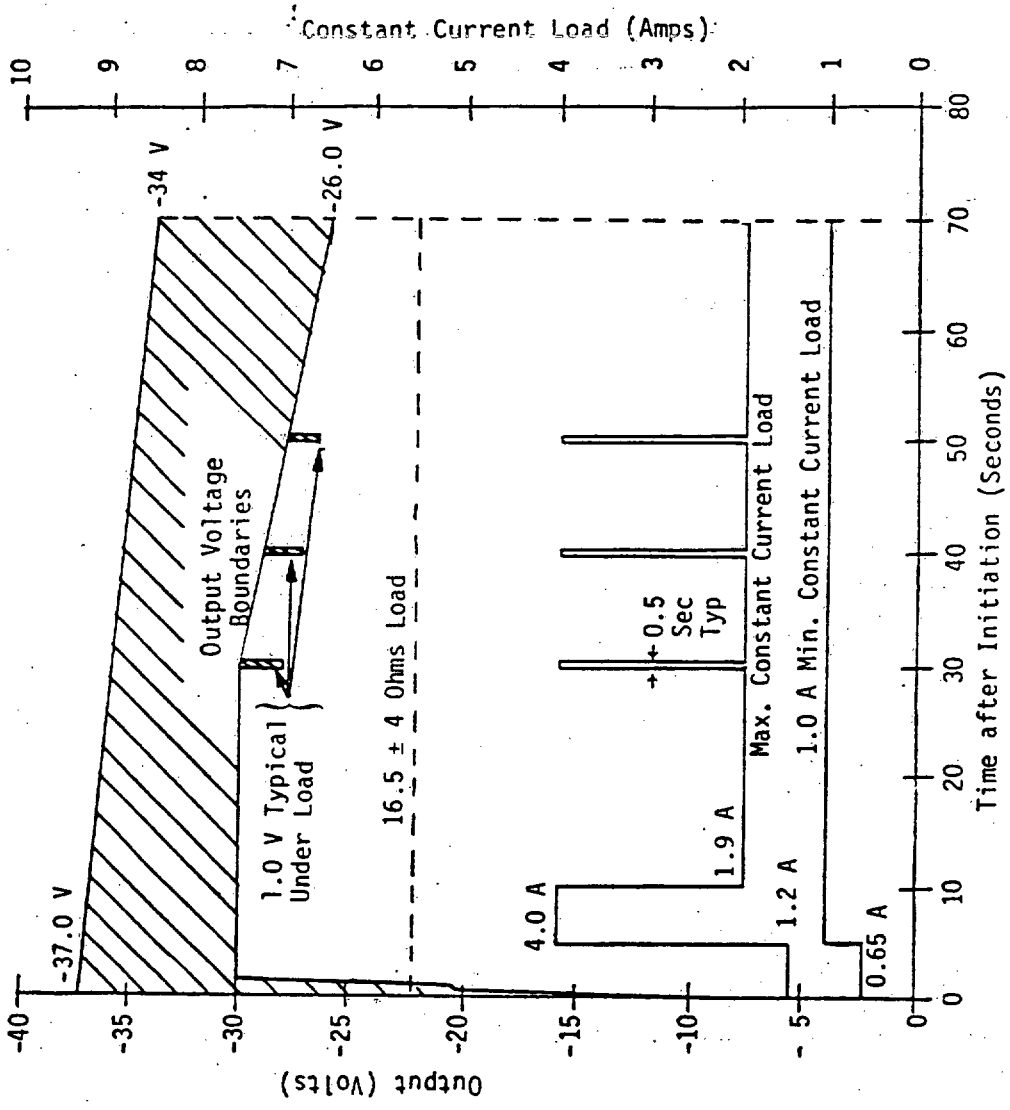


Figure 5. -30 Volt Power Source Voltage and Load Profiles (0 to 70 Seconds)

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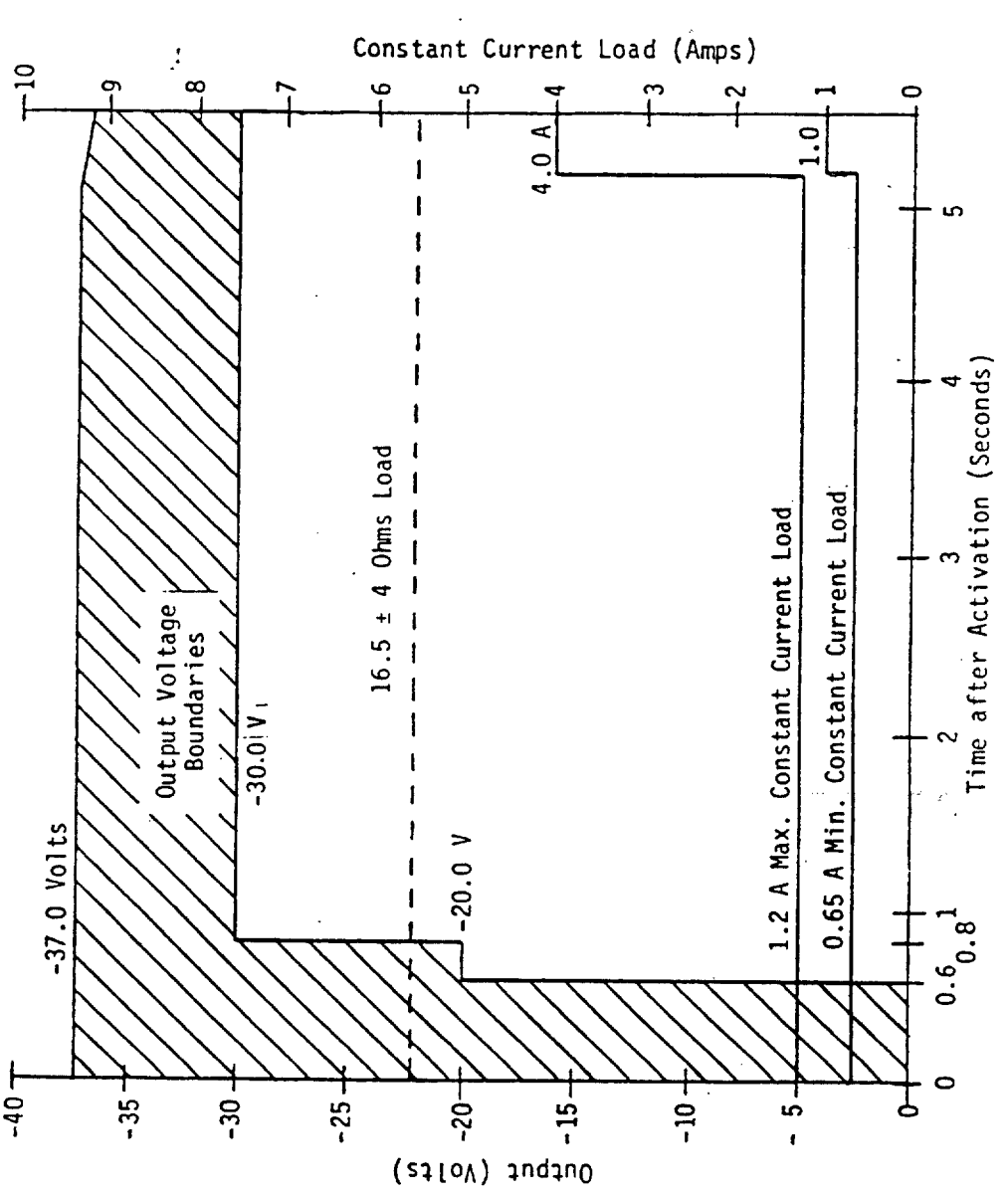


Figure 6. -30 Volt Power Source Voltage and Load Profiles (0 to 5.5 Seconds)

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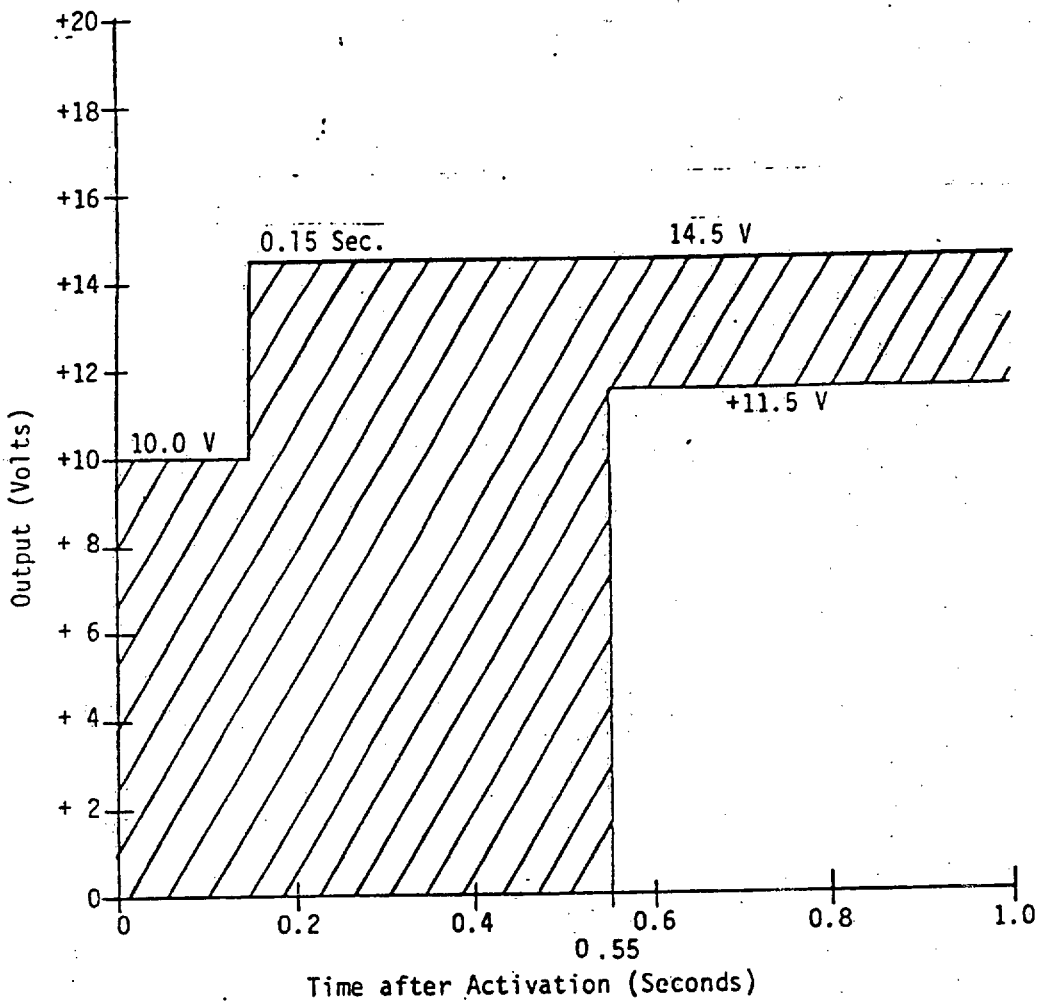


Figure 7. Output Voltage Rise Time Boundaries for +11 Volt Source

NOTE: Refer to paragraph 3.2.1.1.1.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

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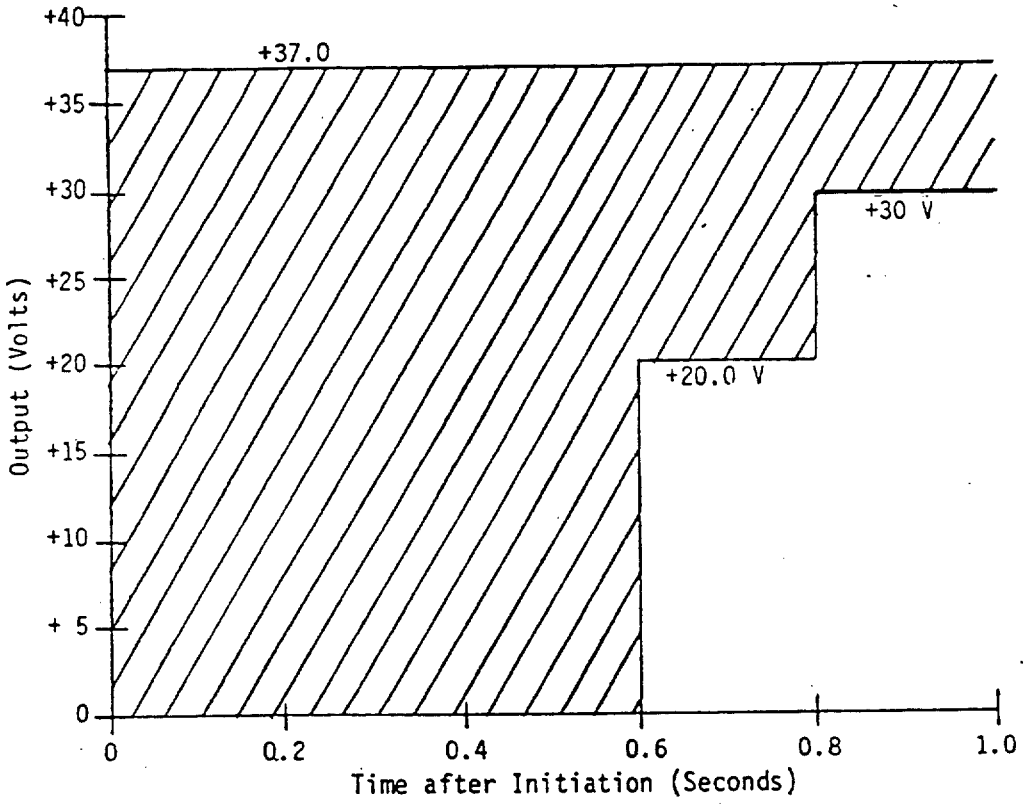


Figure 8. Output Voltage Rise Time Boundaries for the +30 Volt Source

NOTE: Refer to paragraph 3.2.1.1.2.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time

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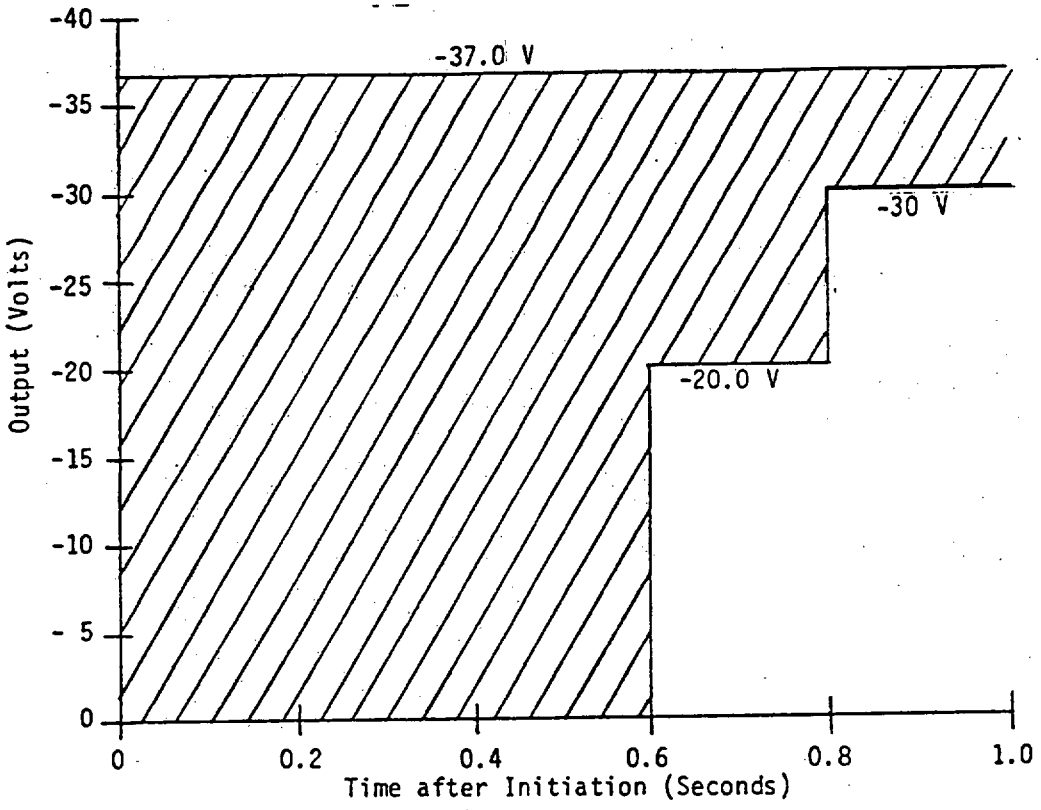


Figure 9. Output Voltage Rise Time Boundaries for the -30 Volt Source

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TABLE I

+ 11 VOLT SOURCE APPLIED SQUIB LOADS

SQUIB LOAD	MAXIMUM LOAD (NOTE 1)	APPLICATION TIME	MINIMUM VOLTAGE	MAXIMUM RECOVERY TIME
	OHMS		VOLTS	
1	31.0	When output voltage reaches 10.0 volts minimum	7.5	Note 2
2	2.5	.210 seconds after first squib is applied	11.5	Note 2
3	1.3	1.5 seconds after first squib is applied	8.0	75 ms
4	1.3	45.0 seconds after first squib is applied	8.0	75 ms

NOTE 1. Squib loads applied for 20.1 $\pm$ 0.1 milliseconds

NOTE 2. Voltage recovery time is 100 milliseconds minus squib load application time.

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Table II
FIRST ARTICLE INSPECTIONS
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2		DRAWING NUMBER
	BATTERY, THERMAL (NOTE 1)			9301200
				NEXT HIGHER ASSEMBLY
				9308000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Terminal strength	100%	3.2.1.4	4.5.2.1
102	Resistance to soldering heat	100%	3.2.1.5	4.5.2.2
103	Polarity	100%	3.2.1.9	4.5.2.10
104	±30V no fire current	100%	3.2.1.2 2.2	4.5.2.7
105	+11V no fire stimulus	100%	3.2.1.2 1.1	4.5.2.8
106	Temperature shock	100%	3.2.2.3	4.5.3.3
107	Transportation vibration	100%	3.2.2.6	4.5.3.5
108	Fragility shock	100%	3.2.2.7	4.5.3.6
109	Case leakage	100%	3.2.1.8	4.5.2.3
110	Launch shock (Note 2)	100%	3.2.2.4	4.5.3.7,4.5.3.8

NOTES:

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Table II
FIRST ARTICLE INSPECTIONS
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2		DRAWING NUMBER 9301200
	BATTERY, THERMAL (NOTE 1)			NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE -INSPECTION METHOD
Major				
111	Overpressure	100%	3.2.2.5	4.5.3.4
112	Case leakage	100%	3.2.1.8	4.5.2.3
113	High temperature operation (Note 3)	100%	3.2.2.1	4.5.3.1, 4.5.2.9
114	Low temperature operation (Note 3)	100%	3.2.2.2	4.5.3.2, 4.5.2.9
Minor	None defined			
	Notes:			
	1) 12 units shall be tested for major items 101 through 105. 8 of those units shall then be selected for testing of major items 106 through 110. The remaining 4 units shall be tested for major items 111 through 114. Testing shall be in the sequence indicated.			
	2) 4 units shall be tested at high g (4.5.3.7) and 4 units shall be tested at low g (4.5.3.8).			
	3) 1/2 of the units from this group shall be tested for major item 113, and the other 1/2 shall be tested for major item 114.			
NOTES:				

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Table III
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BATTERY, THERMAL	SHEET 1 OF 2		DRAWING NUMBER: 9301200
				NEXT HIGHER ASSEMBLY: 9308000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Terminal solderability	0.4%	Dwg.	Dwg.
102	Overall length	0.4%	Dwg.	Dwg.
103	Outside diameter	0.4%	Dwg.	Dwg.
104	Thread dimensions	0.4%	Dwg.	Dwg.
105	Perpendicularity to datum	0.4%	Dwg.	Dwg.
106	True position of case diameter to datum	0.4%	Dwg.	Dwg.
107	Case leakage	100%	3.2.1.8	4.5.2.3
108	Insulation resistance	100%	3.2.1.6	4.5.2.4
109	Dielectric resistance	100%	3.2.1.7	4.5.2.5
110	± 30 V bridge resistance	100%	3.2.1.2.	4.5.2.6
111	Polarity	100%	3.2.1.9	4.5.2.10

NOTES:

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Table III
CONFORMANCE INSPECTIONS, SAMPLING
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2		DRAWING NUMBER
	BATTERY, THERMAL			9301200
				NEXT HIGHER ASSEMBLY
				9308000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Minor				
201	Weight	0.65%	Dwg.	Dwg.
202	Workmanship	0.65%	3.3.2	4.5.5
203	Preparation for delivery	0.65%	5.0	4.5.6

NOTES:

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Table IV
CONFORMANCE INSPECTIONS, LOT ACCEPTANCE TESTS
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: BATTERY, THERMAL	SHEET 1 OF 1		DRAWING NUMBER: 9301200
				NEXT HIGHER ASSEMBLY: 9308000
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined			
Major				
101	Terminal strength	1.0%	3.2.1.4	4.5.2.1
102	Resistance to soldering temperature	1.0%	3.2.1.5	4.5.2.2
103	± 30V no fire current	1.0%	3.2.1.2.2.2	4.5.2.7
104	± 11V no fire stimulus	1.0%	3.2.1.2.1.1	4.5.2.8
105	Transportation vibration	1.0%	3.2.2.6	4.5.3.5
106	High temperature operation (Note 1)	1.0%	3.2.2.1	4.5.3.1
107	Low temperature operation (Note 1)	1.0%	3.2.2.2	4.5.3.2
Minor	None defined			

NOTES:
1) One half of the sample, or the next lowest integer shall be tested per major item 107, remainder shall be tested per major item 106.

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ARMY-AR

PREPARING ACTIVITY
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PROJECT NUMBER;1320-A455

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